

Diophantine Equations — Solutions

Problem 1. (Moldova TST 2021 Problem 2) Prove that if p and q are two prime numbers such that

$$p + p^2 + p^3 + \cdots + p^q = q + q^2 + q^3 + \cdots + q^p,$$

then $p = q$.

Solution. Suppose on the contrary that $p \neq q$. WLOG assume $p < q$. Note that

$$q \mid q + q^2 + q^3 + \cdots + q^p = p + p^2 + p^3 + \cdots + p^q = \frac{p^{q+1} - p}{p - 1}.$$

This implies $q \mid p^{q+1} - p$. As $p \neq q$, we have

$$p^{q+1} - p \equiv p^2 - p = p(p - 1) \pmod{q}$$

by the Fermat little theorem. However, none of p and $p - 1$ is divisible by q since $p < q$. This is a contradiction. So we must have $p = q$. $\square$

Problem 2. (All-Russian MO 2007 Grade 8 Problem 7) For any integer $n > 3$ denote by $n?$ the product of all primes less than n . Solve the equation $n? = 2n + 16$.

Solution. If $n \leq 5$, then $n? \leq 2 \times 3 < 2n + 16$. If $6 \leq n \leq 7$, then $n? = 2 \times 3 \times 5 = 30$. This gives a solution $n = 7$. If $8 \leq n \leq 11$, then $n? = 2 \times 3 \times 5 \times 7 = 210 > 2n + 16$. We claim that there is no solution for $n \geq 12$.

By Bertrand's postulate, there exists a prime p satisfying $\frac{n}{2} < p < n$. Note that $p > \frac{n}{2} > 5$. Therefore, we have

$$2n + 16 = n? \geq 2 \times 3 \times 5 \times p > 15n.$$

This implies $16 > 13n$, contradiction. Therefore, the only solution is $n = 7$. $\square$

Problem 3. (Finnish National High School Mathematics Competition 2018 Problem 5) Solve the Diophantine equation $x^{2018} - y^{2018} = (xy)^{2017}$ where x and y are nonnegative integers.

Solution. Clearly, $x \mid y^{2018}$ and $y \mid x^{2018}$. This shows x and y have the same prime divisors. If one of x and y is 0, so is the other, which gives a solution $x = y = 0$. If x and y are nonzero, consider any prime p which divides x and y . Let $a = v_p(x)$ and $b = v_p(y)$. Then we have

$$v_p(x^{2018}) = 2018a, \quad v_p(y^{2018}) = 2018b, \quad v_p((xy)^{2017}) = 2017(a + b).$$

Note that the smallest two numbers among these are equal. As

$$2017(a + b) > \frac{2018a + 2018b}{2},$$

the smallest two numbers must be $2018a$ and $2018b$. This yields $a = b$. As this holds for every prime divisor p , we must have $x = y$. But then $(xy)^{2017} = x^{2018} - y^{2018} = 0$, which has no nonzero solution. Therefore, the only solution is $x = y = 0$. $\square$

Problem 4. (Thailand MO 2017 Problem 7) Show that no pairs of integers (m, n) satisfy $2560m^2 + 5m + 6 = n^5$.

Solution. Considering modulo 5, we have $1 \equiv n^5 \equiv n \pmod{5}$ by the Fermat little theorem. Let $n = 5k + 1$ where $k \in \mathbb{Z}$. Then we have

$$2560m^2 + 5m + 6 = (5k + 1)^5 \equiv 1 \pmod{25}$$

by the binomial expansion. This yields $25 \mid 10m^2 + 5m + 5$, and hence $5 \mid 2m^2 + m + 1$. But this is impossible since

$$\begin{aligned} 2(0)^2 + (0) + 1 &\equiv 1 \pmod{5}, \\ 2(1)^2 + (1) + 1 &\equiv 4 \pmod{5}, \\ 2(2)^2 + (2) + 1 &\equiv 1 \pmod{5}, \\ 2(-2)^2 + (-2) + 1 &\equiv 2 \pmod{5}, \\ 2(-1)^2 + (-1) + 1 &\equiv 2 \pmod{5}. \end{aligned}$$

This shows there is no solution. □

Problem 5. (India TST 2014 Test 4 Problem 2) Find all positive integers x and y such that $x^{x+y} = y^{3x}$.

Solution. We first prove the following simple fact.

Claim. If $x^a = y^b$ for some positive integers x, y, a, b , then one of x and y is a multiple of the other.

Proof. WLOG assume $(a, b) = 1$. As x^a is a perfect b th power, so is x . Let $x = z^b$ for some positive integer z . Then we have $y = z^a$. Clearly, one of z^b and z^a is a multiple of the other. □

If $x = ky$ for some $k \in \mathbb{N}$, then $(ky)^{k+1} = y^{3k}$. This gives $k^{k+1} = y^{2k-1}$. When $k = 1$, we have $y = 1$. This gives a solution $x = y = 1$. When $k = 2$, we have $y = 2$. This gives a solution $(x, y) = (4, 2)$. When $k = 3$, we have $y^5 = 3^4$, which has no solution. Suppose $k \geq 4$. Note that $(k+1, 2k-1) = (k+1, 3) = 1, 3$.

- If $(k+1, 2k-1) = 3$, then k is a perfect $\frac{2k-1}{3}$ th power. Therefore, we have $k \geq 2^{\frac{2k-1}{3}}$. We prove by induction that $2^{\frac{2k-1}{3}} > k$ for $k \geq 4$. Indeed, for the base case $k = 4$, we have

$$2^{\frac{7}{3}} > 2^2 = 4.$$

Assuming $2^{\frac{2k-1}{3}} > k$, then

$$2^{\frac{2(k+1)-1}{3}} - k = 2^{\frac{2}{3}} 2^{\frac{2k-1}{3}} - k > 2^{\frac{2}{3}} k - k \geq (2^{\frac{2}{3}} - 1)k > 1.$$

This gives the inductive step. Thus, there is no solution for $k \geq 4$.

- If $(k+1, 2k-1) = 1$, then k is a perfect $(2k-1)$ th power. Therefore, $k \geq 2^{2k-1} > 2^{\frac{2k-1}{3}}$. From above, this cannot happen.

If $y = mx$ for some integer $m \geq 2$, then $x^{m+1} = (mx)^3$. This gives $x^{m-2} = m^3$.

- When $m = 2$, there is no solution.
- When $m = 3$, we have $x = 27$. This gives a solution $(x, y) = (27, 81)$.
- When $m = 4$, we have $x = 8$. This gives a solution $(x, y) = (8, 32)$.
- When $m = 5$, we have $x = 5$. This gives a solution $(x, y) = (5, 25)$.

Suppose $m \geq 6$. If m is squarefree, we must have $x \geq m$ since it is clear that x and m share the same prime divisors. But then $m^3 = x^{m-2} \geq m^{m-2} \geq m^4$, which is impossible.

- When $m = 8$, we have $x^6 = 2^9$, which has no solution.
- When $m = 9$, we have $x^7 = 3^6$, which has no solution.
- When $m = 12$, we have $x^{10} = 12^3$, which has no solution.

We now prove that there is no solution for $m \geq 14$. Indeed, as above, we must have $m \geq 2^{\frac{m-2}{3}}$. But then we can prove by induction that $2^{\frac{m-2}{3}} > m$ for $m \geq 14$. For the base case, we have $2^4 = 16 > 14$. For the inductive step, assuming $2^{\frac{m-2}{3}} > m$, then

$$2^{\frac{(m+1)-2}{3}} - m = 2^{\frac{1}{3}} 2^{\frac{m-2}{3}} - m > 2^{\frac{1}{3}} m - m \geq (2^{\frac{1}{3}} - 1)14 > 1.$$

Thus, there is no solution for $m \geq 14$. And the only solutions are

$$(x, y) = (1, 1), (4, 2), (27, 81), (8, 32), (5, 25).$$

□

Problem 6. (Turkey TST for JBMO 2016 Problem 7) Find all pairs (p, q) of prime numbers satisfying

$$p^3 + 7q = q^9 + 5p^2 + 18p.$$

Solution. Rewriting the equation, we obtain

$$f(p) = p^3 - 5p^2 - 18p = q^9 - 7q.$$

Since $q^9 - 7q \geq 2(2^8 - 7)$ and $f(2), f(3) < 0$, we must have $p \geq 5$.

For $x \geq 5$, note that $f'(x) = 3x^2 - 10x - 18 > 0$, so that f is increasing. If $p \leq q^3 + 1$, then

$$f(p) \leq f(q^3 + 1) = q^9 - 2q^6 - 25q^3 - 22 < q^9 - 2q - 25q < q^9 - 7q.$$

If $p \geq q^3 + 2$, then

$$f(p) \geq f(q^3 + 2) = q^9 + q^6 - 26q^3 - 48.$$

In addition, if $q \geq 3$, then

$$q^9 + q^6 - 26q^3 - 48 \geq q^9 + q^3 - 48 \geq q^9 + 9q - 48 \geq q^9 - 7q.$$

Equality holds when $q = 3$ and $p = q^3 + 2 = 29$. So $(p, q) = (29, 3)$ is a solution. Also, this shows there is no solution when $q \geq 5$. Lastly, when $q = 2$, the equation becomes

$$p^3 - 5p^2 - 18p = 498.$$

This shows $p \mid 498 = 2 \times 3 \times 83$. As $p \geq 5$, the only possibility is $p = 83$. This gives $p^2 - 5p - 18 = 6$, i.e. $(p-8)(p+3) = 0$. There is no prime solution. Therefore, the only solution is $(p, q) = (29, 3)$. $\square$

Problem 7. (Cyprus TST 2022 Test 2 Problem 1) Find all pairs of integers (m, n) which satisfy the equation

$$(2n^2 + 5m - 5n - mn)^2 = m^3n.$$

Solution. Noting that $m = n$ is a solution, we factorize as follows.

$$\begin{aligned} & (2n^2 + 5m - 5n - mn)^2 - m^3n \\ &= (2n^2 + 5m - 5n - mn)^2 - (n^2)^2 + (n^4 - m^3n) \\ &= (3n^2 + 5m - 5n - mn)(n^2 + 5m - 5n - mn) + (n^3 - m^3)n \\ &= (3n^2 + 5m - 5n - mn)(n - m)(n - 5) + (n - m)(n^2 + nm + m^2)n \\ &= (n - m)(4n^3 + 10mn - 20n^2 - 25m + 25n + nm^2) \end{aligned}$$

This gives the solution $m = n$. For $m \neq n$, we need

$$nm^2 + 5(2n - 5)m + n(2n - 5)^2 = 0.$$

If $n = 0$, we need $-25m = 0$, so $m = 0$. If $n \neq 0$,

$$\Delta = 25(2n - 5)^2 - 4n^2(2n - 5)^2 = (25 - 4n^2)(2n - 5)^2$$

is a perfect square. This shows $25 - 4n^2 = k^2$ for some nonnegative integer k . Clearly, the only solutions are $(n, k) = (\pm 2, 3)$.

- For $n = 2$, the quadratic equation becomes $2m^2 - 5m + 2 = 0$. The only integer solution is $m = 2$.
- For $n = -2$, the quadratic equation becomes $-2m^2 - 45m - 162 = 0$. The only integer solution is $m = -18$.

Thus, the only solutions are $(m, n) = (t, t), (-18, -2)$ where $t \in \mathbb{Z}$. $\square$

Problem 8. (China TST 2022 Test 4 Problem 4) Find all positive integers a, b, c and prime p satisfying that

$$2^a p^b = (p+2)^c + 1.$$

Solution. Clearly, both sides are even, and hence p is an odd prime. Let $c = 2^k m$ where $k \geq 0$ and $2 \nmid m$.

Suppose c is odd. If c is composite, then it has a proper odd prime divisor q . Note that $(p+2) + 1$, $(p+2)^q + 1$ and $(p+2)^c + 1$ are different divisors of $(p+2)^c + 1$. By Zsigmondy's theorem (and noting $p+2 > 2$), each of them has a distinct prime divisor, contradicting the fact that $(p+2)^c + 1$ only has the prime divisors 2 and p . Therefore, either $c = 1$ or c is a prime.

If $c = 1$, the equation becomes $2^a p^b = p + 3$. As $2^a p^b \geq 2p \geq p + 3$, all equalities hold. The only solution is $(p, a, b) = (3, 1, 1)$.

If c is a prime, then $(p+2) + 1 = p + 3$ is a proper divisor of $(p+2)^c + 1$. Again by Zsigmondy's theorem, $p + 3$ cannot be divisible by $2p$. As it is even, we must have $p + 3 = 2^t$ for some $t \geq 1$. The equation can be rewritten as

$$2^a p^b = (2^t - 1)^c + 1.$$

By the binomial expansion, we find that

$$2^a p^b \equiv c \cdot 2^t \pmod{2^{2t}}.$$

This yields $a = t$. Therefore, we can further rewrite the equation as

$$(p+3)p^b = (p+2)^c + 1. \tag{1}$$

- If $b \leq c - 1$, then

$$(p+3)p^b \leq (p+3)p^{c-1} = (p^2 + 3p)p^{c-2} < (p^2 + 4p + 4)(p+2)^{c-2} < (p+2)^c + 1.$$

- If $b \geq c$, then

$$(p+3)p^b \geq (p+3)p^c = (p+2)p^c + p^c > (p+2)p^c + 1.$$

We claim that $p^c > (p+2)^{c-1}$, which easily implies there is no solution in this case. Indeed, this is equivalent to

$$p > \left(1 + \frac{2}{p}\right)^{c-1}. \tag{2}$$

To obtain an upper bound for c , we consider modulo p in (1). This gives $2^c \equiv -1 \pmod{p}$. Let d be the order of 2 modulo p . Then we have $d \nmid c$ and $d \mid 2c$. Since c is a prime, we can only have $d = 2, 2c$. If $d = 2$, then $p \mid 2^2 - 1 = 3$ so that $p = 3$. This is rejected since $p + 3 \neq 2^t$. Therefore, we must have $d = 2c$. By the Fermat little theorem and the properties of order, we see that $2c \mid p - 1$. Thus, we deduce

$$c \leq \frac{p-1}{2}.$$

Using this, in order to prove (2), it suffices to prove

$$p > \left(1 + \frac{2}{p}\right)^{\frac{p-3}{2}}.$$

It is well-known that the function $f(x) = \left(1 + \frac{1}{x}\right)^x$ is increasing and bounded above by e for $x > 0$ (which can be proved by basic calculus). Therefore, we have

$$\left(1 + \frac{2}{p}\right)^{\frac{p-3}{2}} < f\left(\frac{p}{2}\right) \leq e < p.$$

This proves the assertion.

Suppose c is even. Then we have $(p+2)^c + 1 \equiv 2 \pmod{8}$, and hence $a = 1$. Note that

$$(p+2)^{2^k} + 1 \mid (p+2)^c + 1 = 2p^b.$$

As $(p+2)^{2^k} + 1$ is even and is larger than 2, we must have $2p \mid (p+2)^{2^k} + 1$. This forces $m = 1$ since otherwise $(p+2)^c + 1$ has a prime divisor different from 2 and p by Zsigmondy's theorem. The equation now becomes

$$2p^b = (p+2)^{2^k} + 1.$$

Considering modulo p , we obtain $2^{2^k} \equiv -1 \pmod{p}$. Therefore, the order of 2 modulo p is 2^{k+1} . As above, this implies $2^{k+1} \mid p-1$.

- If $b \leq 2^k - 1$, then

$$2p^b < 2(p+2)^{2^k-1} < (p+2)^{2^k} + 1.$$

- If $b \geq 2^k + 1$, then

$$2p^b \geq 2p^{2^k+1} = (2p-1)p^{2^k} + p^{2^k} > (2p-1)p^{2^k} + 1.$$

We claim that $(2p-1)p^{2^k} > (p+2)^{2^k}$, which easily implies there is no solution in this case. Indeed, this is equivalent to

$$2p-1 > \left(1 + \frac{2}{p}\right)^{2^k}.$$

Recall that $2^{k+1} \mid p-1$. This yields $2^{k+1} \leq p-1$. Therefore, it suffices to prove

$$2p-1 > \left(1 + \frac{2}{p}\right)^{\frac{p-1}{2}}.$$

As above, this holds since

$$\left(1 + \frac{2}{p}\right)^{\frac{p-1}{2}} < f\left(\frac{p}{2}\right) \leq e < 2p-1.$$

This proves the assertion.

- If $b = 2^k$, then the equation becomes

$$2p^{2^k} = (p+2)^{2^k} + 1.$$

Again, we use $2^{k+1} \mid p-1$. If $p-1 \geq 2^{k+2}$, we can follow the previous arguments. We shall prove

$$2 > \left(1 + \frac{2}{p}\right)^{2^k} + \frac{1}{p^{2^k}}.$$

Noting $\frac{1}{p^{2^k}} < 0.2$ and $2^{k+2} \leq p-1$, it suffices to prove

$$1.8 > \left(1 + \frac{2}{p}\right)^{\frac{p-1}{4}}.$$

Indeed, we have

$$\left(1 + \frac{2}{p}\right)^{\frac{p-1}{4}} < f\left(\frac{p}{2}\right)^{\frac{1}{2}} \leq \sqrt{e} < \sqrt{3} < 1.8.$$

It remains to consider the case $p-1 = 2^{k+1}$. The equation becomes

$$2p^{\frac{p-1}{2}} = (p+2)^{\frac{p-1}{2}} + 1.$$

Note that $p = 5$ is a solution since both sides are equal to 50. We show that the right-hand side is larger when $p \geq 7$. It suffices to prove

$$2 < \left(1 + \frac{2}{p}\right)^{\frac{p-1}{2}}.$$

Indeed, by the binomial expansion, we have

$$\left(1 + \frac{2}{p}\right)^{\frac{p-1}{2}} > 1 + \frac{p-1}{2} \cdot \frac{2}{p} + \frac{1}{2} \cdot \frac{p-1}{2} \cdot \frac{p-3}{2} \left(\frac{2}{p}\right)^2 = \frac{5p^2 - 6p + 3}{2p^2} > 2.$$

The last inequality holds since $p \geq 7$.

In conclusion, the only solutions are $(p, a, b, c) = (3, 1, 1, 1), (5, 1, 2, 2)$. □